



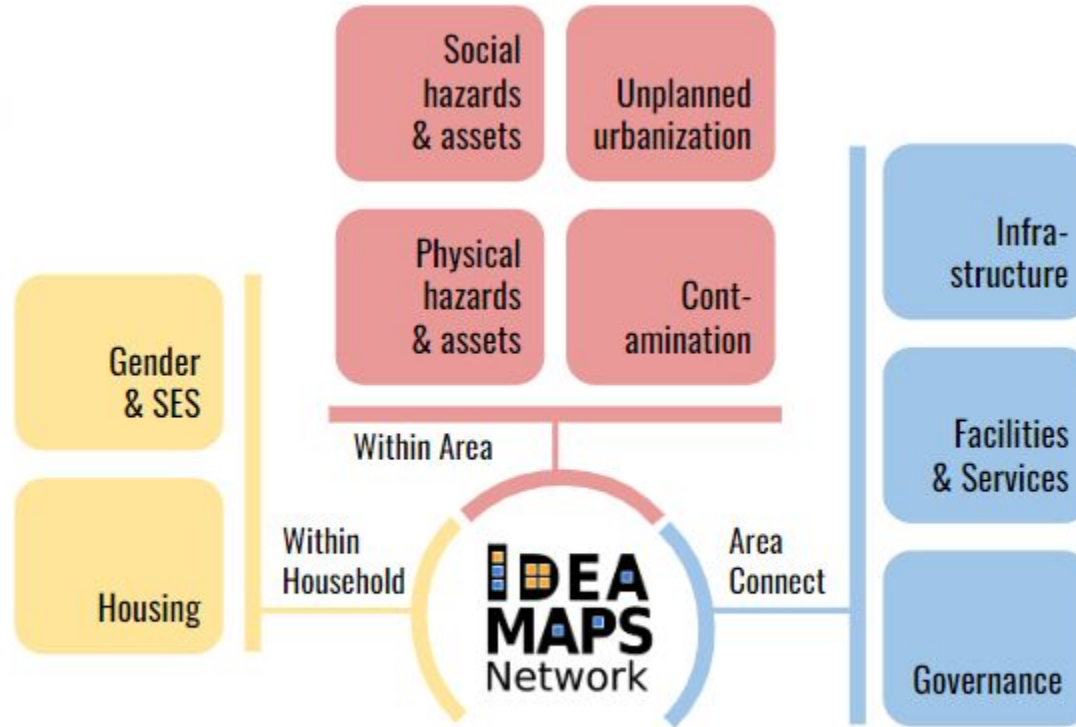
Mapping assets with vulnerable Through OSM and IDEAMAPS

State of the Map SotM 2025

Dundee 2025

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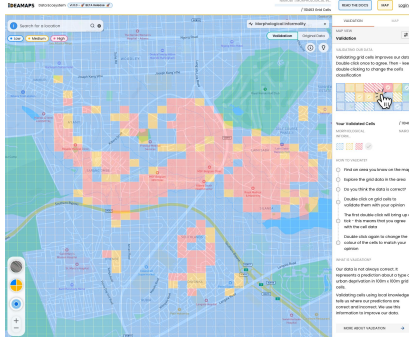
Domains of Urban Deprivation



Domains or Deprivation paper:

<https://doi.org/10.1016/j.compenvurbsys.2022.101770>

Our evolutionary participatory modelling process



1. Codesign and Implementation

Urban model is co-designed with domain experts, local researchers and stakeholders. Model outputs are ingested into the IDEAMAPS Data Ecosystem platform

2. Validation

Model outputs and model specifications are discussed and validated with community members, local government and international stakeholders

3. Improvement

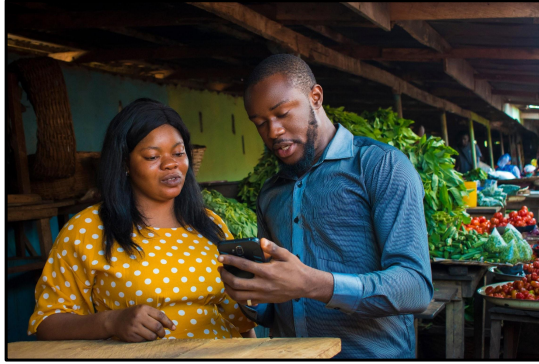
Quantitative data from user interactions in the platform are analysed together with qualitative feedback to determine improvements



Combining “slum” mapping approaches

Field Mapping

using GPS or drawing on printed imagery. Often performed by residents to generate data for planning and advocacy.



Computer models

using AI or machine-learning methods and satellite imagery. Requires training data of slum/non-slum areas.



Census & Survey

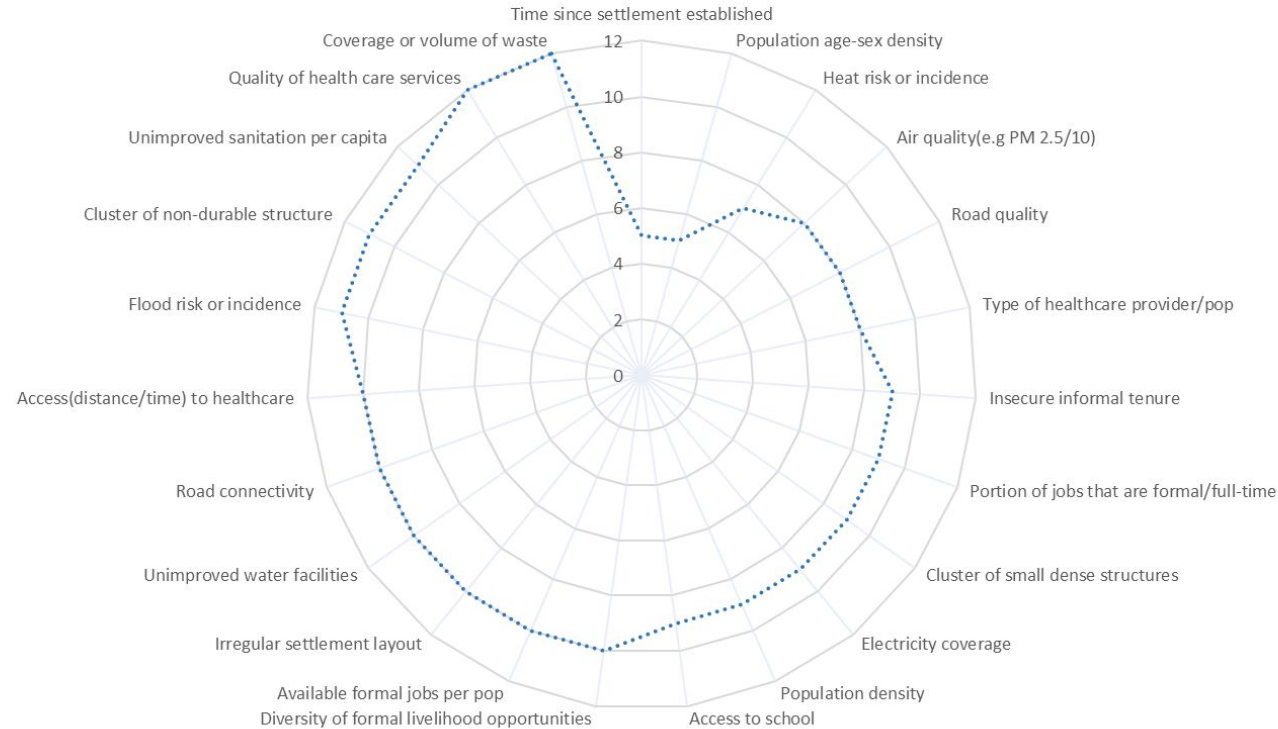
approaches use household-level data to classify “slum” households, then aggregate. An area with >50% “slum” households is a “slum” area.



Digitising imagery is

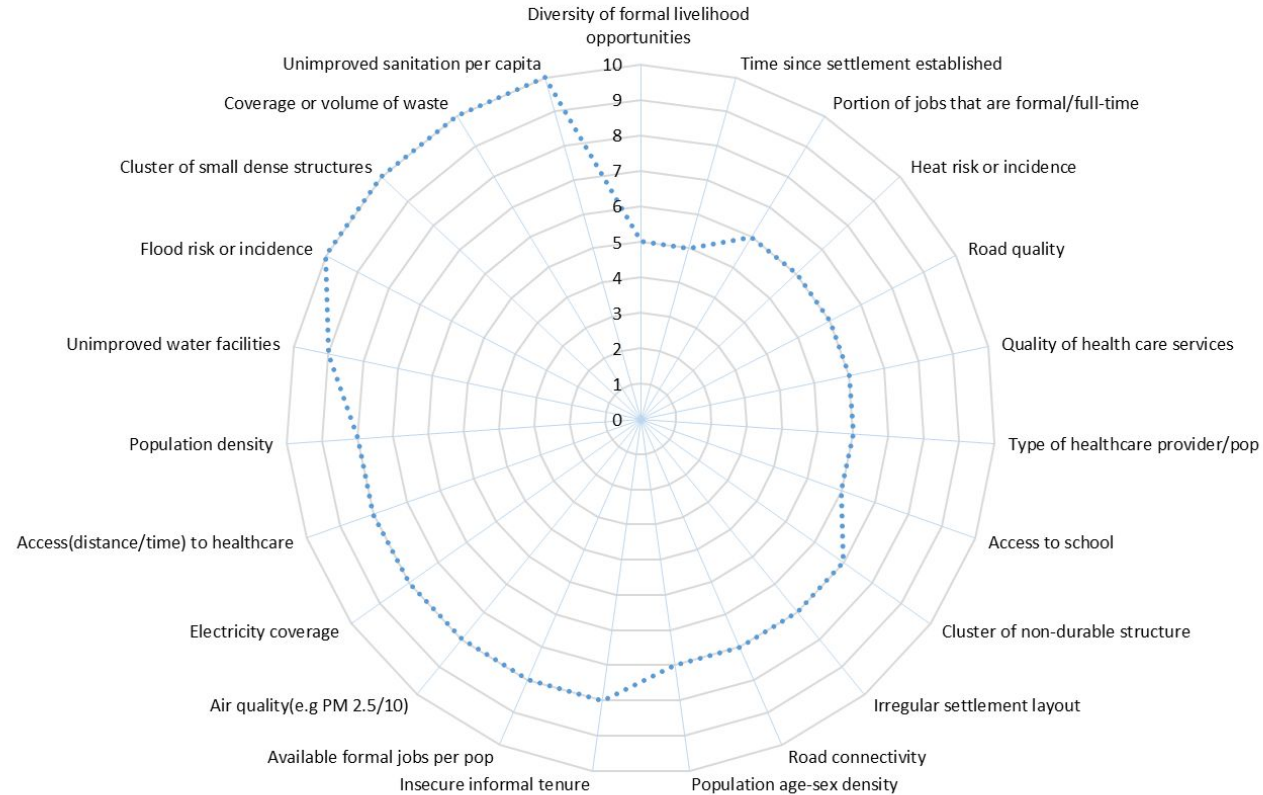
done manually in GIS software, some times by a person unfamiliar with the local context. Digitized imagery is often used to train computer models.

KOROGOCHO SUB-DOMAIN RANKING

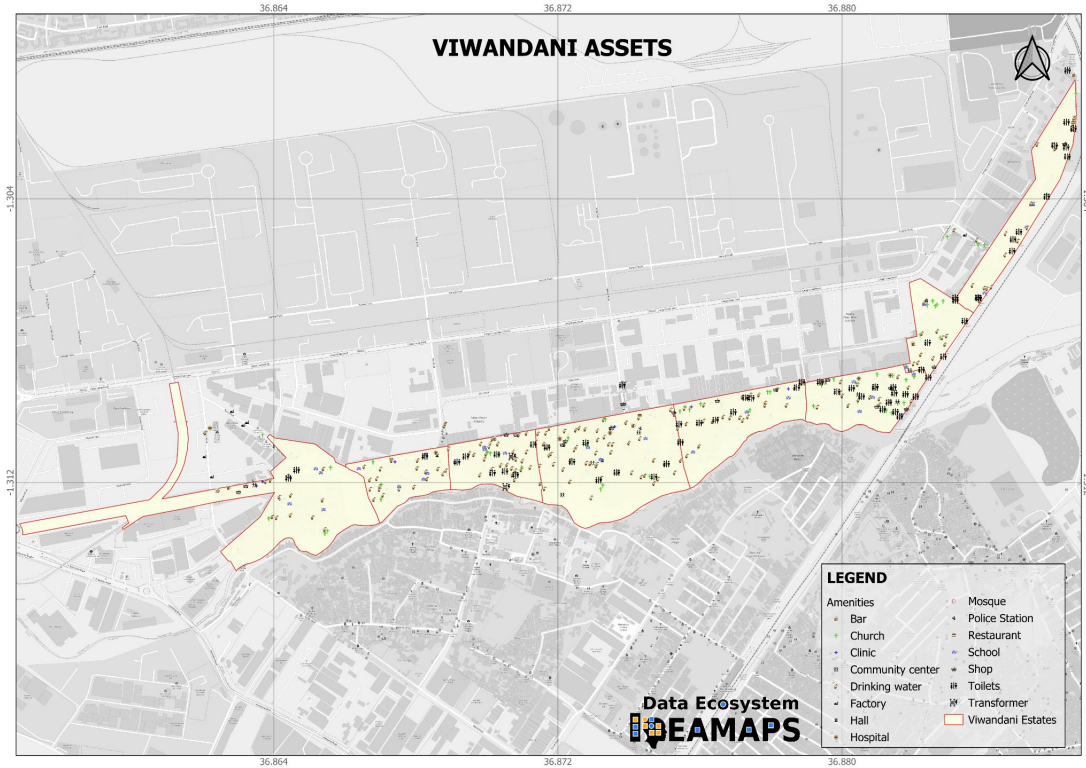
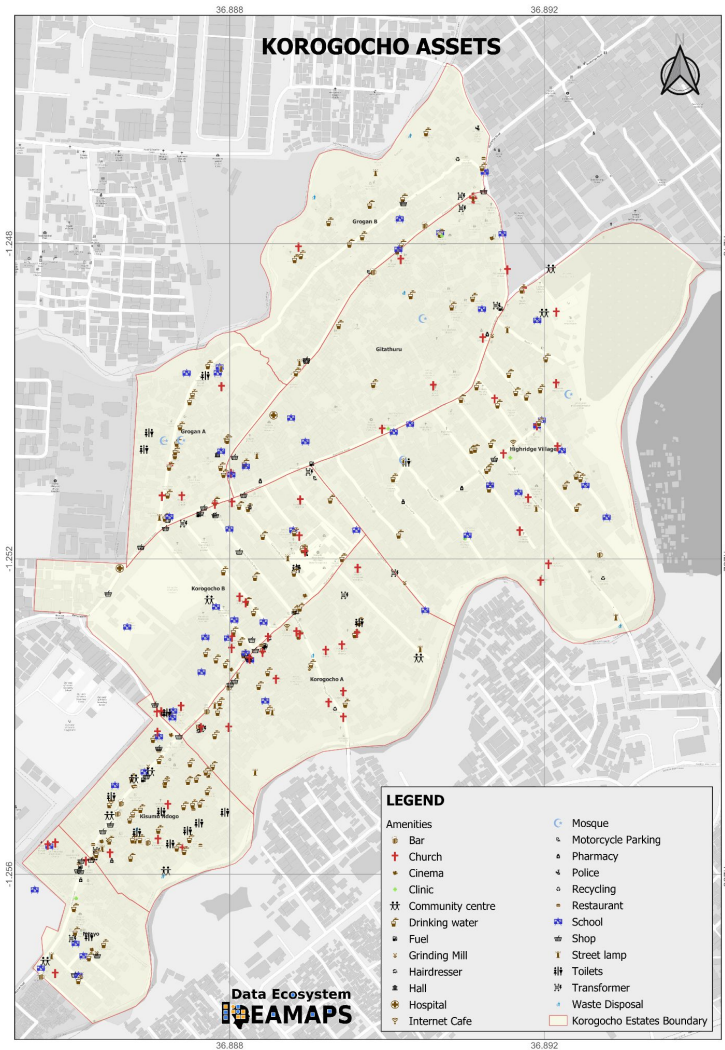


4th largest informal settlement
in Nairobi,
0.97 km² north-east of Nairobi
150,000 residents
250+ dwelling units per Ha

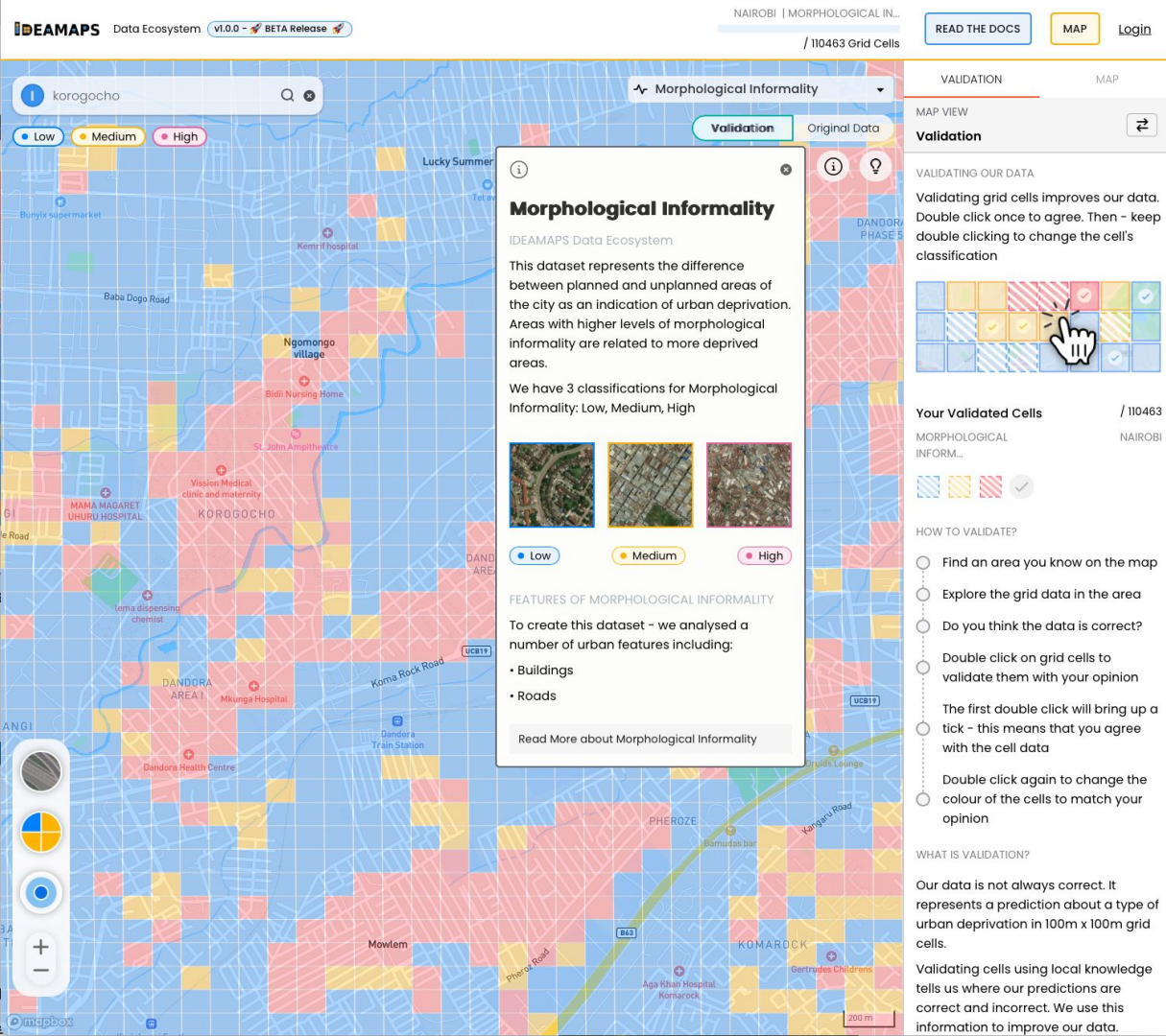
Viwandani Sub-domain ranking



meaning "at the industrial zone" in Kiswahili" in Kiswahili,
South-east part of Nairobi
36,000 residents
14,000 households



Viwandani had 629 points
Korogocho had 443 points



Building a digital platform and process for validating urban data



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